MIL-E-1/1649B
16 February 1979
SUPERSEDING
MIL-E-1/1649A(NAVY)
16 September 1975

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

1/ TYPE 8866

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Any

Weight: 9 ounces (255.2 grams) nominal

ABSOLUTE RATINGS: F = 1,250 MHz

Parameter: Unit:	Ef V (Note 1)	Eb kVdc	Ec1 Vdc	Ec2 Vdc	ib a (Note 2)	Pg2 W	Pp W	tk sec (Min)	tp µs	Du --- (Note 5)	Cooling --- (Note 3)
Maximum: RF power amp class C pulsed:	6.8	3.5	-200	750	5	5	150	60	15	0.02	---
TEST CONDITIONS:	6.3	1.0	---	300	---	---	---	---	---	---	Note 3

GENERAL:

Qualification - Not required

1/ Replaces EIA type 8866 and Admiral Systems Corporation Drawing No. 594A24.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	3.5	4.0	A
1266	Total grid current	4	Ec1/Ib = 300 mAdc	0.65	II	Ic1	---	-5.0	µAdc
1261	Electrode voltage (grid) (cutoff)	-	Ec1/Ib = 10 mAdc	0.65	II	Ec1	-23	-34	Vdc
1261	Electrode voltage (grid)	-	Ec1/Ib = 200 mAdc	0.65	II	Ec1	-8	-15	Vdc
1231	Pulsing emission (sinusoid)	6	tk = 90 sec; ec1 = ec2 = eb = 750 v; t = 90 sec	0.65	II	is	60	---	a
1236	Power oscillation	7	F = 1,075 ±50 MHz; Ebb = 4.0 kVdc; Ec2 = 750 Vdc; Ec1 = -100 Vdc; pd = 1.5 kw (max); tp = 5 to 15 µs; Du = 0.01; t = 5 minutes	0.65	II	po	8	---	kw
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	-	Grid 1 and grid 2 tied together; Cin, anode grounded; Cout, cathode grounded	---	---	Cin Cout	17.0 8.3	24.0 9.5	pF pF
	Quality conformance inspection, part 3								
---	Life test (1)	7	Group D; F = 1,075 ±50 MHz; Ef = 5.5 V; Ebb = 4.0 kVdc; Ec2 = 750 Vdc; Ec1 = -100 Vdc; pd = 1.5 kw; tp = 5 to 15 µs; Du = 0.01; t = 500 hours	---	---	---	---	---	---
---	Life-test end point (1):								
1236	Power oscillation	8		---	---	Δpo t	---	20	%

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NOTES:

1. The cathode of the tube is subjected to considerable back bombardment in ultra-high-frequency service because of transit time effects which raise cathode temperature. The amount of heating due to bombardment is a function of the operating conditions and frequency and shall be compensated for by a reduction of the heater input to prevent overheating of the cathode with resulting short life. Under no condition shall the voltage be reduced below 5.5 volts.

2. A series impedance or crowbar circuit between the anode of the tube and the anode power supply is recommended to limit the peak current under surge conditions to not more than 10 times the maximum rating.

3. A suitable heat sink clamping arrangement shall be provided to limit the anode hub temperature to the specified value. The temperature is measured at the point indicated on figure 1. Airflow required for cooling at sea level through the radiators shall meet the following:

   Heater-cathode seals - - - - - - - - - - - - - - - - 1 CFM (min).

   Screen-grid to control-grid seals- - - - - - - - - 1 CFM (min).

   Incoming air temperature - - - - - - - - - - - - - - 45°C (max).

   Anode hub temperature - - - - - - - - - - - - - - - 250°C (max).

   Seal or ceramic temperature (at any point) - - - - 200°C (max).

   The volumes of cooling air indicated for the various seals is approximate only. Distribution of cooling air will vary with the cavity configuration about the tube. For satisfactory operation, the maximum temperature at any point on the tube shall be below the specified values.

4. This test is to be the first test performed at the conclusion of the holding period.

5. Maximum ratio of on-time to elapsed time during any 7.5 ms period.

6. Applied voltage shall be approximately sinusoidal. The pulse duration is 2.5 to 3.5 µs; prr = 60.

7. The anode voltage power supply shall be terminated in a 2 µF capacitor with a 10-ohm series resistor between power supply and tube cavity.

8. At the conclusion of life test, the power output under the life test conditions specified shall be not less than 7.5 kw (useful), and Δpo_t shall not exceed 20 percent; Δpo_t is the percentage decrease in power output from that achieved under optimum conditions when the tube was initially started on life test, after allowing a period not to exceed 1 hour for the tube to stabilize in operation.

Custodians:

Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

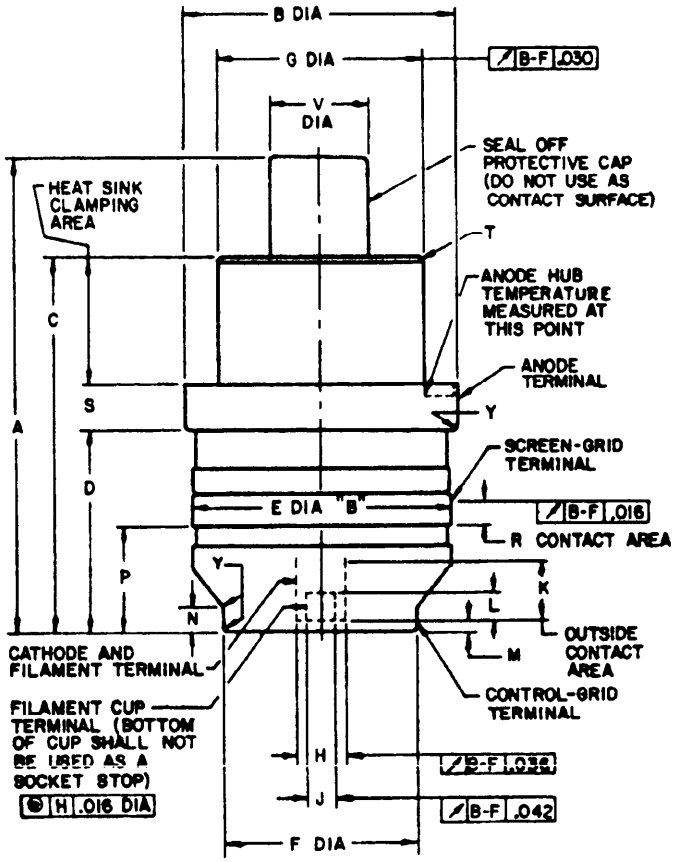
User activities
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3214)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.875 (73.03)	3.125 (53.98)
B	1.740 (44.20)	1.760 (44.70)
C	2.313	2.438
D	1.250	1.313
E	1.620	1.640
F	1.302	1.322
G	1.320	1.330
H	0.320	0.328
J	0.181	0.191
K	0.375	
L	0.188	
M	0.063	0.125
N	0.219	
P	0.656	0.719
R	0.109	
S	0.250	0.313
T		0.030
V		0.625
Y		0.047

FIGURE 1. Outline drawing of electron tube type 8866.

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